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Controlled stopping for a compactor machine

Abstract

A compactor machine may include a frame, a compaction member attached to the frame, and a controller attached to the frame. The controller may be configured to detect, while the compactor machine is traveling on a paving material mat to perform compacting, an event that is to cause the compactor machine to perform a stop. The controller may be configured to identify, based on detecting the event, a zone that the compactor machine is to use to perform the stop. The controller may be configured to cause the compactor machine to travel to the zone to perform the stop.

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Background/Summary

TECHNICAL FIELD

(1) The present disclosure relates generally to compactor machines and, for example, to controlled stopping for a compactor machine.

BACKGROUND

(2) Compaction of a surface material, such as soil or asphalt, can improve strength and stability of the surface. In a paving context, a paving machine distributes hot paving material, such as asphalt, over a surface, and a mobile compactor machine follows the paving machine to compact the material to a desired density and obtain an acceptable surface finish. Generally, a consistent density and smooth surface finish may be achieved by the compactor machine travelling over the hot paving material at a relatively constant speed.

(3) However, sometimes during a compacting operation, the compactor machine may need to stop while on hot paving material. This may be particularly prevalent for a compactor machine operating autonomously, where the compactor machine may be configured to stop if an object is detected in a path of the compactor machine or if the compactor machine loses a positioning signal for navigation or loses communications with nearby machines. As a result of the compactor machine stopping while on hot paving material, a surface of the paving material may be deformed or damaged, thereby requiring reworking or repair of the surface that consumes machine hours, increases machine wear, and/or increases fuel consumption.
(4) U.S. Pat. No. 8,477,021 (the '021 patent) discloses a worksite proximity warning system to avoid vehicle collisions. The '021 patent states that the proximity warning system is configured to provide both low level and high level alarms to an operator when an obstacle approaches or enters a safe zone, and when the machine approaches and enters a hazard zone. While avoiding obstacles is a reason that a compactor machine may stop on hot paving material, the '021 patent does not address how to avoid deformation or damage of a surface of a paving material mat due to the compactor machine stopping.
(5) The controller of the present disclosure solves one or more of the problems set forth above and/or other problems in the art.
SUMMARY
(6) A compactor machine may include a frame, a compaction member attached to the frame, and a controller attached to the frame. The controller may be configured to detect, while the compactor machine is traveling on a paving material mat to perform compacting, an event that is to cause the compactor machine to perform a stop. The controller may be configured to identify, based on detecting the event, a zone that the compactor machine is to use to perform the stop. The controller may be configured to cause the compactor machine to travel to the zone to perform the stop.
(7) A method may include detecting, by a controller of a compactor machine, while the compactor machine is traveling on a paving material mat to perform compacting, an event that is to cause the compactor machine to perform a stop. The method may include determining, by the controller and based on at least one of temperature sensor data or density sensor data relating to the paving material mat, whether the compactor machine is to perform the stop at a location of the compactor machine at a time that the event is detected. The method may include performing, by the controller, one or more actions based on a determination that the compactor machine is not to perform the stop at the location of the compactor machine.
(8) A controller for a compactor machine may include one or more memories and one or more processors coupled to the one or more memories. The one or more processors may be configured to detect, while the compactor machine is traveling on a paving material mat to perform compacting, an event that is to cause the compactor machine to perform a stop. The one or more processors may be configured to identify, based on detecting the event, a zone that the compactor machine is to use to perform the stop. The one or more processors may be configured to cause the compactor machine to travel to the zone to perform the stop.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of an example machine described herein.
(2) FIG. **2** is a diagram of an example associated with controlled stopping for a compactor machine.
(3) FIG. **3** is a flowchart of an example process associated with controlled stopping for a compactor machine.
DETAILED DESCRIPTION
(4) FIG. **1** is a perspective view of an example machine **100** described herein. While in FIG. **1** the

machine **100** is depicted as a compactor machine, the machine **100** may be another type of machine configured to perform compaction of a ground material. The machine **100** may be an asphalt compactor machine (e.g., a self-propelled, double-drum compactor machine), a vibratory drum compactor machine, or the like, which may be used to compact various materials, such as soil and/or asphalt, among other examples.

(5) The machine **100** has at least one compaction member, such as a compaction drum. For example, as shown, the machine **100** has a front compaction drum **102** and a back compaction drum **104**. The compaction drums **102**, **104** are a set of ground-engaging members that provide ground engagement of the machine **100** at surfaces **102′**, **104′** of the compaction drums **102**, **104**, respectively. The surfaces **102′**, **104′** may include cylindrical surfaces that form exteriors of shells of the compaction drums **102**, **104**, respectively. As the machine **100** passes over a mat of paving material, the surfaces **102′**, **104′** roll against the paving material and provide compaction forces to the paving material due to a weight of the machine **100**. One or more of the compaction drums **102**, **104** may include a vibratory component configured to cause the compaction drums **102**, **104** to vibrate, thereby further facilitating compaction. In some examples, the machine **100** may include one or more other ground-engaging members, such as one or more wheels and/or one or more tracks, in addition or alternatively to the front compaction drum **102** or the back compaction drum **104**.

(6) The machine **100** includes an operator station **106** equipped with various systems and/or mechanisms for control of the operation of the machine **100**. For example, the operator station **106** may include a drive system control **108** (shown as a shift lever) and/or a steering system control **110** (shown as a steering wheel). A steering system of the machine **100** may include the steering system control **110**, a steering column (e.g., connected to the steering system control **110**), a steering actuator (e.g., a steering cylinder for power steering), and/or a steering linkage assembly (e.g., that connects the steering system control **110** or the steering column to ground engagement members, such as the compaction drums **102**, **104**, via a plurality of linkage members, such as rods). The operator station **106** may also include a display **112** that provides a graphical user interface for operating the machine **100**.

(7) The machine **100** includes an engine **114** and a generator **116** coupled with the engine **114**. The engine **114** and the generator **116** are attached to a frame **118** of the machine **100**. The generator **116** may serve as an electrical power source for various onboard systems and components of the machine **100**. The engine **114** may include any type of engine (e.g., an internal combustion engine, a gasoline engine, a diesel engine, a gaseous fuel engine, or the like). The engine **114** is configured to drive movement of the machine **100** (e.g., via compaction drums **102**, **104**) and other components of the machine **100**, such as the generator **116**. In some examples, the machine **100** may include an electric motor and an electrical power storage device (e.g., a battery), and/or a fuel cell power source, additionally or alternatively to the engine **114**. The compaction members **102**, **104** can be connected to the frame **118**. The machine **100** also includes a braking system **120** configured to receive operator input to decrease or arrest a speed of the machine **100**.

(8) The machine **100** includes a controller **122** (e.g., an electronic control module (ECM)) supported by, or otherwise attached to, the frame **118**. The controller **122** may include one or more memories **124** and one or more processors **126** communicatively coupled to the one or more memories **124**. A processor **126** may include a central processing unit, a graphics processing unit, a microprocessor, a controller, a microcontroller, a digital signal processor, a field-programmable gate array, an application-specific integrated circuit, and/or another type of processing component. The processor **126** may be implemented in hardware, firmware, or a combination of hardware and software. The processor **126** may be capable of being programmed to perform one or more operations or processes described elsewhere herein. A memory **124** may include volatile and/or nonvolatile memory. For example, the memory **124** may include random access memory (RAM), read only memory (ROM), a hard disk drive, and/or another type of memory (e.g., a flash memory,

a magnetic memory, and/or an optical memory). The memory **124** may be a non-transitory computer-readable medium. The memory **124** may store information, one or more instructions, and/or software (e.g., one or more software applications) related to the operation of the controller **122**.

(9) As indicated above, FIG. **1** is provided as an example. Other examples may differ from what is described in connection with FIG. **1**.

(10) FIG. **2** is a diagram of an example **200** associated with controlled stopping for a compactor machine described herein. As shown, a plurality of machines may perform work at a worksite **205** to condition a ground surface **210**. The plurality of machines may include the machine **100** and one or more additional work machines, such as one or more paving machines and/or one or more compactor machines. For example, a paving machine may perform a paving operation to distribute a paving material mat (e.g., an asphalt mat) over the ground surface **210**, and the machine **100** may follow the paving machine to compact the paving material. In some implementations, the plurality of machines may use machine-to-machine communication to exchange information with each other, such as location data and/or sensor data. In some implementations, the machine **100** may operate at the worksite **205** in an autonomous driving mode, in which propulsion, steering, and braking is controlled autonomously by the machine **100**. Alternatively, driving operations of the machine **100**, such as propulsion, steering, and braking, may be controlled manually by an operator of the machine **100**, who may be stationed on the machine **100** or at a remote console.

(11) As the machine **100** travels on the paving material mat distributed by the paving machine, the controller **122** may generate one or more maps. A map may include a collection of data relating to locations of the worksite **205** (e.g., such data can be, but does not have to be, expressed in a visual format). The controller **122** may generate a time map indicating times and locations associated with travel of the machine **100** on the paving material mat. As an example, as the machine **100** travels on the paving material mat (e.g., to perform compacting), the controller **122** may periodically detect a location of the machine **100** (e.g., using a global navigation satellite system (GNSS)), and the controller **122** may record the location in association with a timestamp, thereby generating a time map indicating a chronological list of times and locations of the machine **100** on the paving material mat.

(12) Additionally, or alternatively, the controller **122** may generate a heat map indicating temperatures of the paving material mat at various locations. The controller **122** may generate the heat map based on temperature sensor data relating to the paving material mat collected by the machine **100** and/or collected by one or more work machines that are at the worksite **205** with the machine **100** (e.g., the machine **100** may receive, via machine-to-machine communication, the temperature sensor data collected by the work machines). For example, as the machine **100** travels on the paving material mat (e.g., to perform compacting), the controller **122** may periodically detect a location of the machine **100** (e.g., using a GNSS) and a temperature of the paving material mat, and the controller **122** may record the location in association with the temperature, thereby generating a heat map indicating a list of temperatures and locations of the paving material mat. The heat map may also indicate respective times associated with the temperatures, which may indicate times that temperature sensor data was collected for each temperature of the heat map. In some implementations, the controller **122** may update temperatures of the heat map based on the times (e.g., amounts of elapsed time from the times) and a known cooling rate of the paving material (e.g., at a particular ambient temperature and/or ambient humidity).

(13) Additionally, or alternatively, the controller **122** may generate a density map indicating densities of the paving material mat at various locations. The controller **122** may generate the density map based on density sensor data relating to the paving material mat collected by the machine **100** and/or collected by one or more work machines that are at the worksite **205** with the machine **100** (e.g., the machine **100** may receive, via machine-to-machine communication, the density sensor data collected by the work machines). For example, as the machine **100** (or another

work machine) travels on the paving material mat (e.g., to perform compacting), the controller **122** may periodically detect a location of the machine **100** (e.g., using a GNSS) and a density of the paving material mat, and the controller **122** may record the location in association with the density, thereby generating a density map indicating a list of densities and locations of the paving material mat. The controller **122** may update the density map as new density sensor data is collected by the machine **100** (or another work machine) in connection with additional compaction passes. The density sensor data may be sensor data that directly indicates density measurements, or sensor data that indicates other types of measurements that can be used to estimate density.

(14) In some implementations, one or more of the other work machines may also generate a time map, a heat map, and/or a density map. Accordingly, the controller **122** may receive one or more maps from another work machine.

(15) While the machine **100** is traveling on the paving material mat, the controller **122** may detect an event that is to cause the machine **100** to perform a stop (e.g., to stop traveling). The event may be that the machine **100** has lost a communication connection (e.g., machine-to-machine communication) with one or more work machines that are at the worksite **205** with the machine **100**. Additionally, or alternatively, the event may be that the machine **100** has lost a communication connection with a supervisory system (e.g., a back office device) for the machine **100** and/or a remote control device for the machine **100**. For example, without the communication connection, the machine **100** may not receive location data from the work machines or may not receive instructions or commands from the supervisory system or the remote control device, and the machine **100** should stop to avoid a collision. The controller **122** may detect that the machine **100** has lost the communication connection with the one or more work machines, the supervisory system, and/or the remote control device based on detecting an absence of received communications (e.g., when such communications are scheduled or configured to be received) over a threshold time period and/or based on detecting an absence of acknowledgements being received, for communications transmitted by the machine **100**, over a threshold time period.

(16) The event may be that the machine **100** has lost a positioning signal used for navigation (e.g., one or more satellite navigation signals). For example, without the positioning signal, the machine **100** may be unable to detect a location of the machine **100**, and the machine **100** should stop to avoid a collision. The controller **122** may detect that the machine **100** has lost the positioning signal based on detecting an absence of the positioning signal (e.g., due to signal blockage, due to receiver malfunction, or due to a loss of time synchronization) over a threshold time period.

(17) The event may be detection of an obstruction (e.g., an object or a person in a path of the machine **100**) within a particular distance (e.g., 20 feet, 15 feet, or the like) of the machine **100**. For example, based on the obstruction being within the particular distance of the machine **100**, the machine **100** should stop to avoid a collision. The controller **122** may detect the obstruction using an autonomous driving system of the machine **100**, which may include one or more cameras, one or more lidar systems, or the like. For example, the controller **122** may detect the obstruction based on processing camera data and/or lidar data using computer vision techniques or other machine learning techniques.

(18) The event may be the paving machine (that the machine **100** is following) stopping. For example, when the paving machine stops, the machine **100** also should stop to avoid a collision with the paving machine. The controller **122** may detect the paving machine stopping using machine-to-machine communication with the paving machine (e.g., the machine **100** may receive an indication from the paving machine, such as location data associated with the paving machine, indicating that the paving machine is stopped). Additionally, or alternatively, the controller **122** may detect the paving machine stopping using an autonomous driving system of the machine **100**, in a similar manner as described above.

(19) The event may be a machine fault of the machine **100**, such as an electrical system failure or an electrical component failure, a mechanical system failure or a mechanical component failure, or

a hydraulic system failure or a hydraulic component failure. For example, the controller **122** may detect a machine fault based on losing communication with a system or a component and/or based on receiving a fault notification from a system or a component, among other examples. Based on detecting the machine fault, the machine **100** should stop to permit repair of the machine **100**. The event may be a battery level or a fluid level (e.g., a fuel level, a diesel exhaust fluid level, a water level, or the like) dropping below a threshold. For example, the controller **122** may monitor the battery level or the fluid level to detect when the battery level or the fluid level drops below the threshold. Accordingly, the machine **100** should stop to permit recharging of the battery or refilling of the fluid.

(20) Based on detecting the event that is to cause the machine **100** to perform the stop, the controller **122** may determine whether the machine **100** may perform the stop at a location of machine **100** at the time the event is detected (e.g., a current location of the machine **100**). For example, the controller **122** may determine whether the machine **100** may perform the stop at the location based on temperature sensor data and/or density sensor data relating to the paving material mat (e.g., at the location). For example, if the temperature sensor data indicates a temperature of the paving material mat at the location that exceeds a threshold (e.g., the temperature is too high), and/or if the density sensor data indicates a density of the paving material mat at the location that is below a threshold (e.g., the density is too low), then the controller **122** may determine that the machine **100** is not to perform the stop at the location.

(21) The controller **122** may perform one or more actions based on a determination that the machine **100** is not to perform the stop at the location. For example, the controller **122** may cause an indicator (e.g., an indicator light, an indicator on display **112**, or the like) to indicate that the machine **100** is not to perform the stop at the location of the machine **100** (e.g., the indicator may turn red, may start blinking, or the like). When the machine **100** is manually operated, this may prompt an operator to drive the machine **100** elsewhere. While the machine **100** is in transit, the controller **122** may continue to determine whether the machine **100** may perform the stop at a current location (e.g., based on temperature sensor data and/or density sensor data, as described above), and the controller **122** may activate the indicator accordingly. For example, based on a determination that the machine **100** may perform the stop at a current location of the machine **100**, the controller **122** may cause the indicator to indicate that the machine **100** can perform the stop at the current location (e.g., the indicator may turn green, may stop blinking, or the like).

(22) Additionally, or alternatively, based on detecting the event and/or based on a determination that the machine **100** is not to perform the stop at the location, the controller **122** may identify a zone **215** (e.g., one or more zones **215**) that the machine **100** is to use to perform the stop (which can be referred to as a “holding zone,” a “stopping zone,” a “parking zone,” or the like). The zone **215** may be an area of the worksite **205** (e.g., that is smaller than a total area of the worksite **205**). In some examples, the zone **215** may be a location of the worksite **205**, designated for stopping, that is configured for the machine **100**. For example, an operator of the machine **100** or a supervisor of the worksite **205** may designate the location for stopping (e.g., by providing an input to the machine **100** or to a back office device that may control the machine **100**). In some examples, to identify the zone **215**, the controller **122** may determine the zone **215** based on temperature sensor data and/or density sensor data (e.g., collected by the machine **100** and/or by one or more other work machines) relating to the paving material mat. For example, based on a determination that the machine **100** is not to perform the stop at the location, the machine **100** may begin to travel elsewhere (whether by manual control or by autonomous control via the controller **122**), and as the machine is traveling, the controller **122** may determine whether the machine **100** may perform the stop at a current location of the machine **100** (e.g., based on temperature sensor data and/or density sensor data, as described above). Accordingly, the machine **100** may determine the zone **215** to encompass a location where the controller **122** has determined that the machine **100** may perform the stop.

(23) In some examples, to identify the zone **215**, the controller **122** may determine the zone **215** based on the time map generated by the controller **122**. For example, the controller **122** may identify a location of the time map associated with a time from which a threshold amount of time has elapsed (e.g., 1 hour, 2 hours, or the like), and the controller **122** may use the location as the zone **215** (e.g., because the location has likely cooled enough to allow for stopping of the machine **100**). Additionally, or alternatively, to identify the zone **215**, the controller **122** may determine the zone **215** based on the heat map generated by the controller **122**. For example, the controller **122** may identify a location of the heat map associated with a temperature that is below a threshold, and the controller **122** may use the location as the zone **215**. Additionally, or alternatively, to identify the zone **215**, the controller **122** may determine the zone **215** based on the density map generated by the controller **122**. For example, the controller **122** may identify a location of the density map associated with a density that exceeds a threshold, and the controller **122** may use the location as the zone **215**.
(24) The controller **122** may cause the machine **100** to travel to the zone **215**, identified by the controller **122**, to perform the stop (e.g., causing the machine **100** to travel to the zone **215** may be an action performed by the controller **122**, as described above). The controller **122** may cause the machine **100** to travel to the zone **215** by generating a control signal for an autonomous driving system of the machine **100**. For example, the control signal may identify geographic coordinates (e.g., latitude and longitude coordinates) associated with the zone **215**. In some examples, the controller **122** may detect, based on location data associated with the machine **100**, that the machine **100** is in the zone **215**, and the controller **122** may cause the machine **100** to perform the stop in the zone **215** (e.g., by generating a control signal for the braking system **120**).
(25) After the machine **100** is stopped in the zone **215**, the controller **122** may detect an additional event indicating that the machine **100** can leave the zone **215**. For example, the additional event may include the machine **100** regaining a communication connection with one or more work machines that are at the worksite **205** with the machine **100** (e.g., if the event was the machine **100** losing the communication connection), the machine **100** regaining a positioning signal for navigation (e.g., if the event was the machine **100** losing the positioning signal), detection of clearing of an obstruction and/or receiving an indication of clearing of the obstruction (e.g., if the event was detection of the obstruction), and/or the paving machine (that the compactor machine is following) resuming movement (e.g., if the event was the paving machine stopping). Based on detecting the additional event, the controller **122** may cause the machine **100** to return to the paving material mat (e.g., to a last-worked location on the paving material mat). If the controller **122** detects a subsequent event that is to cause the machine **100** to perform a stop, then the controller **122** may cause the machine **100** to return to the zone **215** (or a different zone **215**), in a similar manner as described above, and so forth.
(26) As indicated above, FIG. **2** is provided as an example. Other examples may differ from what is described in connection with FIG. **2**.
(27) FIG. **3** is a flowchart of an example process **300** associated with controlled stopping for a compactor machine. One or more process blocks of FIG. **3** may be performed by a controller (e.g., controller **122**). Additionally, or alternatively, one or more process blocks of FIG. **3** may be performed by another device or a group of devices separate from or including the controller, such as another device or component that is internal or external to machine **100**.
(28) As shown in FIG. **3**, process **300** may include detecting, while a compactor machine is traveling on a paving material mat to perform compacting, an event that is to cause the compactor machine to perform a stop (block **310**). For example, the controller may detect the event, as described above. The compactor machine may be operating in an autonomous driving mode. The event may be that the compactor machine has lost a communication connection with one or more work machines that are at a worksite with the compactor machine, with a supervisory system for the compactor machine, or with a remote control device for the compactor machine. The event may

be that the compactor machine has lost a positioning signal used for navigation. The event may be the detection of an obstruction within a particular distance of the compactor machine. The event may be a paving machine, that the compactor machine is following, stopping. The event may be a machine fault of the compactor machine. The event may be a battery level or a fluid level of the compactor machine dropping below a threshold.
(29) As further shown in FIG. **3**, process **300** may include determining, based on at least one of temperature sensor data or density sensor data relating to the paving material mat, whether the compactor machine is to perform the stop at a location of the compactor machine at a time that the event is detected (block **320**). For example, the controller may determine whether the compactor machine is to perform the stop, as described above.
(30) As further shown in FIG. **3**, process **300** may include performing one or more actions based on a determination that the compactor machine is not to perform the stop at the location of the compactor machine (block **330**). For example, the controller may perform one or more actions, as described above. In some examples, performing the one or more actions may include causing an indicator to indicate that the compactor machine is not to perform the stop at the location of the compactor machine.
(31) Process **300** may include identifying, based on detecting the event, a zone that the compactor machine is to use to perform the stop. In some examples, performing the one or more actions may include causing the compactor machine to travel to the zone to perform the stop. Causing the compactor machine to travel to the zone may include generating a control signal for an autonomous driving system of the compactor machine, where the control signal identifies geographic coordinates associated with the zone. The zone may be a location, designated for stopping, that is configured for the compactor machine. Alternatively, identifying the zone may include generating a map indicating times and locations associated with travel of the compactor machine on the paving material mat, and determining the zone based on the map. Additionally, or alternatively, identifying the zone may include determining the zone based on at least one of temperature sensor data or density sensor data relating to the paving material mat. For example, identifying the zone may include generating at least one of a heat map or a density map of the paving material mat based on sensor data collected by at least one of the compactor machine or one or more work machines that are at a worksite with the compactor machine, and determining the zone based on the at least one of the heat map or the density map. Process **300** may include detecting, based on location data associated with the compactor machine, that the compactor machine is in the zone, and causing the compactor machine to perform the stop in the zone.
(32) Although FIG. **3** shows example blocks of process **300**, in some implementations, process **300** may include additional blocks, fewer blocks, different blocks, or differently arranged blocks than those depicted in FIG. **3**. Additionally, or alternatively, two or more of the blocks of process **300** may be performed in parallel.
INDUSTRIAL APPLICABILITY
(33) The controller described herein may be used with any mobile machine configured to perform compaction of a material, such as a paving material (e.g., asphalt). For example, the controller may be used with a compactor machine equipped with one or more compaction drums that are configured to roll against a paving material and provide compaction forces to the paving material. In some examples, the controller may be used with a compactor machine that performs compacting operations autonomously.
(34) Sometimes during a compacting operation, a compactor machine may need to stop while on hot paving material. For example, a compactor machine operating autonomously may be configured to stop if an object is detected in a path of the compactor machine or if the compactor machine loses a positioning signal for navigation or loses communications with nearby machines. As a result, a surface of the paving material may be deformed or damaged, thereby requiring reworking or repair of the surface that consumes machine hours, increases machine wear, and/or

increases fuel consumption.
(35) The controller described herein is useful for reducing deformation or damage to paving material that is being worked by a compactor machine. In particular, the controller may detect an event that is to cause the compactor machine to perform a stop, but rather than immediately stopping when the event is detected, the controller may cause the compactor machine to travel to a particular stopping zone to perform the stop. The stopping zone may be a location associated with paving material that has cooled enough and/or that is dense enough for the compactor machine to stop on without causing deformation or damage to the paving material. Additionally, or alternatively, the stopping zone may be a location of a worksite that is other than a surface being paved. In this way, the controller may reduce deformation or damage of a surface being paved, thereby reducing or preventing the need to rework or repair the surface. Accordingly, the controller may conserve machine hours, reduce machine wear, and/or conserve fuel.
(36) The foregoing disclosure provides illustration and description, but is not intended to be exhaustive or to limit the implementations to the precise forms disclosed. Modifications and variations may be made in light of the above disclosure or may be acquired from practice of the implementations. Furthermore, any of the implementations described herein may be combined unless the foregoing disclosure expressly provides a reason that one or more implementations cannot be combined. Even though particular combinations of features are recited in the claims and/or disclosed in the specification, these combinations are not intended to limit the disclosure of various implementations. Although each dependent claim listed below may directly depend on only one claim, the disclosure of various implementations includes each dependent claim in combination with every other claim in the claim set.
(37) As used herein, “a,” “an,” and “a set” are intended to include one or more items, and may be used interchangeably with “one or more.” The term “attached” is used in the broadest sense to encompass any connection between two structures, whether direct or indirect (that is, through the use of one or more intervening elements or bonding materials). Further, as used herein, the article “the” is intended to include one or more items referenced in connection with the article “the” and may be used interchangeably with “the one or more.” Further, the phrase “based on” is intended to mean “based, at least in part, on” unless explicitly stated otherwise. Also, as used herein, the term “or” is intended to be inclusive when used in a series and may be used interchangeably with “and/or,” unless explicitly stated otherwise (e.g., if used in combination with “either” or “only one of”).

Claims

1. A compactor machine, comprising: a frame; a compaction member attached to the frame; and a controller, attached to the frame, configured to: generate at least one of a heat map or a density map of the paving material mat based on sensor data collected by at least one of the compactor machine or one or more work machines that are at a worksite with the compactor machine; detect, while the compactor machine is traveling on a paving material mat to perform compacting, an event that is to cause the compactor machine to perform a stop; identify, based on detecting the event, a zone that the compactor machine is to use to perform the stop, wherein the zone is determined based on the at least one of the heat map or the density map; and cause the compactor machine to travel to the zone to perform the stop.
2. The compactor machine of claim 1, wherein the compaction member comprises a front compaction drum and a back compaction drum.
3. The compactor machine of claim 1, wherein the event is the compactor machine has lost a communication connection with one or more work machines that are at a worksite with the compactor machine, with a supervisory system for the compactor machine, or with a remote control device for the compactor machine.

4. The compactor machine of claim 1, wherein the event is the compactor machine has lost a positioning signal used for navigation.
5. The compactor machine of claim 1, wherein the event is detection of an obstruction within a particular distance of the compactor machine.
6. The compactor machine of claim 1, wherein the event is a paving machine, that the compactor machine is following, stopping.
7. The compactor machine of claim 1, wherein the zone is a location, designated for stopping, that is configured for the compactor machine.
8. The compactor machine of claim 1, wherein the controller, to identify the zone, is configured to: generate a map indicating times and locations associated with travel of the compactor machine on the paving material mat; and determine the zone based on the map.
9. The compactor machine of claim 1, wherein the controller, to cause the compactor machine to travel to the zone, is configured to: generate a control signal for an autonomous driving system of the compactor machine, the control signal identifying geographic coordinates associated with the zone.
10. A method, comprising: generating, by a controller of a compactor machine, at least one of a heat map or a density map of a paving material mat based on sensor data collected by at least one of the compactor machine or one or more work machines that are at a worksite with the compactor machine; determining a zone based on the at least one of the heat map or the density map; detecting, by the controller of the compactor machine, while the compactor machine is traveling on the paving material mat to perform compacting, an event that is to cause the compactor machine to perform a stop; determining, by the controller and based on at least one of the heat map or the density map relating to the paving material mat, whether the compactor machine is to perform the stop at a location of the compactor machine at a time that the event is detected; and performing, by the controller, one or more actions based on a determination that the compactor machine is not to perform the stop at the location of the compactor machine.
11. The method of claim 10, further comprising: identifying, by the controller and based on detecting the event, the zone that the compactor machine is to use to perform the stop.
12. The method of claim 11, wherein performing the one or more actions comprises: causing, by the controller, the compactor machine to travel to the zone to perform the stop.
13. The method of claim 10, wherein performing the one or more actions comprises: causing, by the controller, an indicator to indicate that the compactor machine is not to perform the stop at the location of the compactor machine.
14. The method of claim 10, wherein the compactor machine is operating in an autonomous driving mode.
15. A controller for a compactor machine, comprising: one or more memories; and one or more processors, coupled to the one or more memories, configured to: detect, while the compactor machine is traveling on a paving material mat to perform compacting, an event that is to cause the compactor machine to perform a stop, wherein the event is at least one of: the compactor machine has lost a communication connection with one or more work machines that are at a worksite with the compactor machine, with a supervisory system for the compactor machine, or with a remote control device for the compactor machine; identify, based on detecting the event, a zone that the compactor machine is to use to perform the stop; and cause the compactor machine to travel to the zone to perform the stop.
16. The controller of claim 15, wherein the one or more processors, to identify the zone, are configured to: determine the zone based on at least one of temperature sensor data or density sensor data relating to the paving material mat.
17. The controller of claim 15, wherein the one or more processors, to identify the zone, are configured to: generate at least one of a heat map or a density map of the paving material mat based on sensor data collected by at least one of the compactor machine or one or more work machines

that are at a worksite with the compactor machine; and determine the zone based on the at least one of the heat map or the density map.

18. The controller of claim 15, wherein the one or more processors are further configured to: detect, based on location data associated with the compactor machine, that the compactor machine is in the zone; and cause the compactor machine to perform the stop in the zone.